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July 02 2026

Agile Full Stack Web Development - Ecommerce Project Proposal

A Winnipeg based company called Superex has contracted me to build an e-commerce site for their rapidly growing outerwear business. They were founded in 2008 with roughly 40 employees. They currently sell technical outdoor gear and a premium lifestyle line. They currently have two Winnipeg retail stores and they sell wholesale through stores like MEC and Bass Pro Shops in other provinces. They have no online store, and demand for their product has skyrocketed, they intend to sell both outdoor gear and their premium lifestyle line online. Their target demographics are adults who like to do outdoor/hiking activities and young adults who want to wear clothing that has a premium feel and luxury style.

Tables

USERS table holds the customer accounts along with their address details. This table stores the users so that every order can be linked back to the customer who placed it, and passwords are saved in a hashed and salted format.

PROVINCES table has all 13 provinces and territories with their names, abbreviations and tax rates, online orders are only available in Canada. This table is needed because the checkout has to calculate GST, PST and HST based on where the customer lives.

PRODUCTS table is the store's catalogue and holds everything about a product like the name, description, price, stock and sale status. The created_at and updated_at columns are also used for the new and recently updated product filters.

CATEGORIES table stores the product categories like Shell Jackets or Lifestyle. We use this table so that customers can browse the store by category and also search within a specific category.

CATEGORIES_PRODUCTS is a join table that connects products to categories. It is needed because one product can belong to more than one category.

ORDERS table saves one row every time a customer completes a checkout. It stores the order status, subtotal, the tax amounts that were charged, the grand total, and the Stripe IDs so the order can be matched with the payment in the payment processor.

ORDER_ITEMS table stores the products inside each order along with the quantity and the unit price at the time of purchase. The unit price is saved here so that changing a product's price later will not affect what past orders show.

PAGES table holds the content for the About and Contact pages. Pages tables are used primarily by the admin so that they can edit these pages from a web form in the dashboard instead of changing the code.

ADMINS table stores the staff accounts for the admin dashboard. It is kept separate from the users table on purpose so a customer account can never get admin access.

Associations

- A province has many users, and a user belongs to a province.
- A province has many orders, and every order belongs to one province because that is the province the taxes were calculated with.
- A user has many orders, and each order belongs to one user. This is a one-to-many relationship.
- An order has many order_items, and each order_item belongs to one order. An order must have at least one item in it.
- A product has many order_items, meaning orders and products are connected through the order_items table.
- A category has many categories_products entries, and each categories_products entry belongs to one category.
- A product has many categories_products entries, and each categories_products entry belongs to one product. This and the previous bullet together is what makes products and categories a many-to-many relationship.
- The pages and admins tables are not connected to anything, and this is on purpose.

